Dhamane Maithili Sunil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.20
K.V.N Naik	Class XII (Maharashtra State Board)	2020	83.54%
Kilbil ST. Joseph's High School	Class X (Maharashtra State Board)	2018	94.60%

PROJECTS

DriveVQA: Multi-view Visual Question Answering for Autonomous Driving

- Architected a multimodal Visual Question Answering system for autonomous driving that interprets 6-camera view scenes and answers natural-language questions spanning perception and planning, enabling **interpretable post-hoc analysis** of driving decisions.
- Engineered a lightweight vision-language architecture integrating **ViT-B/32** for visual encoding, **gated multi-view feature fusion**, and fine-tuned **T5-Base** for answer generation, using a **two-stage training strategy** to balance computational efficiency and interpretability.
- Achieved **BLEU-4: 53.09, METEOR: 36.19 , ROUGE-L: 66.79 , CIDEr: 2.79** on the DriveLM-nuScenes benchmark, validating the model's ability to generate accurate, contextually grounded answers for driving scene understanding.

Explainable Predictive Maintenance System

- Built an end-to-end predictive maintenance system using the **AI4I 2020 dataset** to predict machine failures from sensor readings, incorporating feature engineering, imbalanced-class learning, and a **Streamlit dashboard** for interpretable failure-risk predictions and anomaly insights.
- Engineered domain-specific features and benchmarked Logistic Regression, Decision Tree, **Random Forest, and XGBoost**. Handled severe **class imbalance** with SMOTE applied inside the training pipeline. Added **SHAP explanations** and **Isolation Forest anomaly detection** for interpretability.
- Achieved **F1 score of 0.76**, with **Recall 0.82** and **PR-AUC 0.88**, so most real failures are caught while false alarms stay controlled.

ClauseFinder: Automated Clause Extraction from Commercial Contracts

- Developed a **legal contract clause-extraction system** using the CUAD dataset (510 contracts, 41 clause types) to identify key clauses in lengthy commercial contracts and return structured JSON for human review.
- Benchmarked three approaches: a fine-tuned **encoder-based extractive QA model**, a **prompted open-source LLM**, and **LoRA fine-tuning** of the same LLM. Handled long contracts with **overlapping chunking** and **contract-level** train/validation/test **splits** to prevent data leakage.
- Evaluated using **token-level precision, recall, and F1-score** per clause category, alongside **inference latency**, **cost per document**, and learning curves across training-set sizes.

M.TECH ONGOING THESIS

Multimodal Visual Grounding on Synthetic Aperture Radar Images

- Developing a **multimodal deep learning** system for SAR visual grounding, which **localizes objects** in radar imagery from natural-language queries.
- Implemented a **transformer-based** grounding pipeline using **BERT** (text) and **ResNet-50** (image) with cross-modal attention and a deformable-attention decoder.
- Added configurable **speckle-denoising filters** (Lee, Enhanced Lee, Non-Local Means) to the preprocessing stage.
- Exploring the impact of denoising through a controlled ablation and investigating image-text encoders for stronger cross-modal alignment.

SKILLS

- **Programming Language**: C/C++, Python, SQL

- **Machine Learning/ Deep Learning**: Classification, Regression, Anomaly Detection, CNNs, RNN, Transformers
- **Libraries**: PyTorch, Scikit-learn, XGBoost, Hugging Face, Pandas, Numpy
- **MLOps**: Streamlit, FastAPI, Docker

POSITIONS OF RESPONSIBILITY

Organizer Pre-AI Impact Summit '26

- **Anchored** for an audience of **~500 attendees**, including national and international delegates, while **coordinating** summit **events** to ensure smooth execution and audience engagement.

Co-Ordinator, Public Relations and Networking- TEDxIITHyderabad

- Owned end-to-end speaker coordination, from identifying and shortlisting speakers to managing outreach, follow-ups, and travel and accommodation arrangements, ensuring a seamless experience for invited speakers.

Secretary, Rotaract Club of Nashik Smart City

- Coordinated **7 club committees**, facilitated cross-club collaborations, and represented Rotaract Club of Nashik Smart City at **state and national level meets**, strengthening coordination across club initiatives and inter-club engagement.